

Archivara Agent: Exact Falsification of Cellular-Automaton Routes to Arithmetic Takeya Certificates

Archivara Research Pipeline / Codex CLI

March 19, 2026

Abstract. We study a FrontierMath-style finite-certificate benchmark for the arithmetic Takeya problem: find a legal verifier-side certificate $(X; d_i; f_i; T; R)$ with score at most 1.675, ideally using a cellular-automaton-inspired local grammar. The present repository state does not support a positive cellular-automaton claim. Instead, it supports an exact benchmark-and-falsification paper. We formalize the low-height grid compiler used throughout the saved experiments and prove that the repository verifier is sound and complete for this specialization. Two exact obstruction theorems then kill the headline positive routes: a one-seed theorem eliminates the frozen H1 corridor pilot, and a passive-middle row-quotient theorem eliminates the height-3 lift of the typed-interface family. The remaining route audits are negative by exact computation. H4 typed residue interfaces produce 0 forcing width-4 words out of 39 canonical candidates, so composition never begins. The width-8 complexity ledger shows that every matched rerun at fixed budget is empty; the saved exploratory 29/14 witness survives only outside that ledger. The H5 affine-core / nonlinear-shell audit finds 0 forcing shell words across 117 canonical checks, while matched direct search on low-height asymmetric families finds exact certificates of score 13/7, 13/7, and 14/7. The H6 phase-coded schedules, exhaustive controller enumeration over 75,276 programs, and 27-word metastability scan yield no advantage over the same direct frontier. The best surviving exact objects are therefore direct low-height asymmetric certificates at score 13/7 and 14/7, still well above the benchmark target.

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1 Introduction

The geometric Kakeya conjecture asks whether every bounded set in $\mathbb{R}^d$ that contains a unit line segment in every direction must have full Hausdorff and Minkowski dimension d . Katz and Tao introduced an arithmetic projection formalism strong enough to imply geometric Kakeya lower bounds, and Green and Ruzsa later gave a clean modern account of the arithmetic Kakeya conjecture in that finite-set language [Katz and Tao, 1999, Green and Ruzsa, 2019]. In the notation used throughout this paper, the conjectural endpoint is AK(1): finitely many oblique projections should control the anti-diagonal projection with exponent 1, not merely the trivial exponent 2. Via Bourgain’s slicing argument and subsequent Szemerédi-type transfer machinery, progress on AK(α) yields dimension lower bounds for geometric Kakeya sets [Bourgain, 1999, Leng et al., 2024].

The benchmark studied in this repository packages that program into an exact finite search problem. Given the verifier-side forcing formalism from the task statement, one asks for a legal certificate $(X; d_i; f_i; T; R)$ with score

$$\frac{m(G) + |R|}{n(G) - |T|} \leq 1.675,$$

where $m(G)$ counts nonzero edge labels, $n(G)$ counts vertices, R is the initial singleton-generator set, and T is the initial solved set [Epoch AI, 2025, Archivara Research Lab, 2026n]. The user request that launched the repo was explicit: *solve this using cellular automata*. The present manuscript shows why that slogan has to be rejected, at least for the audited routes saved in the repository.

The difficulty is not only mathematical; it is epistemic. Bootstrap percolation, critical cellular automata, abelian networks, chip-firing systems, and decoder-style local rules all offer natural metaphors for forcing and propagation [Bollobas et al., 2015, Hartarsky and Mezei, 2020, Bond and Levine, 2016a,b,c, Sipser and Spielman, 1996, Hemenway et al., 2015, Kubica and Preskill, 2019]. But in arithmetic Kakeya the source of truth is not a rollout picture. It is the original forcing semantics: a certificate counts only if an exact verifier can derive anti-diagonal singleton relations by legal integer-span operations. The repository therefore treats cellular-automaton language only as a proposal mechanism for candidate certificates, never as evidence in its own right [Archivara Research Lab, 2026o,b].

At the current repository endpoint, every tested cellular-automaton-positive lane has either failed exactly or collapsed to parity with matched direct search. The Phase 4 corridor pilot already killed the frozen H1 route by a one-seed obstruction and found only small direct controls at score 2.0 [Archivara Research Lab, 2026g,f,i]. Phase 6 pushed the negative story further. H4 typed residue interfaces yield 0 forcing width-4 words out of 39 canonical candidates and therefore never reach a nontrivial width-8 composition test [Archivara Research Lab, 2026j]. The width-8 complexity ledger shows that every matched rerun with fixed budget is empty, so the saved exploratory 29/14 witness is not part of a valid matched frontier [Archivara Research Lab, 2026d]. The H5 shell audit finds no forcing shell word across 117 canonical checks, while matched direct search on low-height asymmetric families produces exact certificates of score 13/7, 13/7, and 14/7 [Archivara Research Lab, 2026k]. The H6 schedule family, exhaustive controller search, and metastability summaries recover the same best direct object rather than a new mechanism [Archivara Research Lab, 2026l,c,m]. The honest endpoint is therefore negative: the best surviving exact rows are direct certificates, not cellular-automaton-derived ones.

Our contributions are the following.

1. We formalize the low-height grid specialization used throughout the repository and prove that

the exact rational-span verifier is sound and complete for this family. This theorem covers every saved Phase 4 and Phase 6 search row in the paper.

2. We prove a general one-seed obstruction theorem and apply it to the frozen H1 corridor pilot, recovering the repository’s exact Phase 4 route kill from first principles [Archivara Research Lab, 2026g,h].
3. We prove a stronger passive-middle row-quotient obstruction theorem for the height-3 lift of the typed-interface family. Unlike the one-seed theorem, it allows arbitrary boundary seeds and arbitrary width inside that family [Archivara Research Lab, 2026e,c].
4. We synthesize the saved Phase 6 route audits: H4 interfaces fail before composition, the matched width-8 ledger stays empty, H5 bounded shell words lose to direct search, and H6 schedules, controller search, and metastability show no lift over the direct frontier [Archivara Research Lab, 2026j,d,k,l,c,m].
5. We identify the actual positive frontier currently present in the repository: direct low-height asymmetric width-4 certificates at scores 13/7, 13/7, and 14/7, all still above the target 1.675 [Archivara Research Lab, 2026k].
6. We position the work honestly relative to prior art: it is an exact benchmark-and-falsification paper inside the existing arithmetic-Keakeya certificate setting, not a new theorem line, not a new formulation, and not a validated cellular-automaton mechanism [Green and Ruzsa, 2019, Cowen-Breen et al., 2020, Pohoata and Zakharov, 2024, Tao, 2025, Archivara Research Lab, 2026o,q].

Figure 1 summarizes the repository pipeline at the current endpoint. Section 2 explains the novelty boundary. Section 3 restates the arithmetic benchmark and defines the grid specialization. Section 4 describes the exact verifier, the direct-search routines, and the route-specific audits. Section 5 proves the compiler/verifier theorems and the two obstruction theorems. Sections 6 and 7 present the saved experiments. Sections 8 and 9 close with limitations, open questions, and the final evidence-based interpretation.

2 Related Work

2.1 Arithmetic Keakeya and Neighboring Theorem Lines

The closest mathematical line is the arithmetic Keakeya program itself. Katz and Tao introduced the projection framework that motivates the benchmark studied here [Katz and Tao, 1999]. Green and Ruzsa gave a modern formulation and several equivalent perspectives on the conjecture [Green and Ruzsa, 2019]. Cowen-Breen, Karangozishvili, Varadarajan, and Wang developed related pattern problems inside the same finite-certificate universe [Cowen-Breen et al., 2020]. Pohoata and Zakharov extended the theorem-level landscape with a generalized arithmetic Keakeya statement [Pohoata and Zakharov, 2024]. Tao’s analysis of bounded-many-slopes sets and rational complexity is especially relevant because it warns that tiny fixed- X or low-complexity families can look active without meaningfully approaching the real conjectural frontier [Tao, 2025]. Lemm’s counterexample line is an additional caution against over-trusting overly symmetric or homogeneous search spaces [Lemm, 2015].

This paper is not another theorem in that line. It proves no new arithmetic Kakeya exponent, no new implication, and no new equivalent formulation. The mathematical object is always the original verifier-side certificate. The contribution is operational and negative: a collection of exact compiler, verifier, obstruction, and benchmark results that eliminate several cellular-automaton-adjacent routes inside the existing certificate setting [Archivara Research Lab, 2026o,b].

2.2 Bootstrap Growth, Abelian Networks, and Decoder Overlap

The natural nearby languages all come from local propagation. Bootstrap percolation and critical cellular automata suggest droplet, threshold, or interface heuristics [Bollobas et al., 2015, Hartarsky and Mezei, 2020]. Abelian networks and chip-firing suggest commutative invariant packages based on rank, torsion, and recurrent-state structure [Bond and Levine, 2016a,b,c]. Decoder and expander-code ideas suggest local correction, message passing, or sweep-style transport heuristics [Sipser and Spielman, 1996, Hemenway et al., 2015, Kubica and Preskill, 2019]. The repository tested all three types of overlap language, but only in the limited role justified by the verifier. H2 used abelian-style descriptors only as a screening package; H4 and H5 froze local interface and gate alphabets and asked whether those alphabets produced legal exact certificates; H6 treated phase schedules, controllers, and metastability metrics as finite audited proposal banks [Archivara Research Lab, 2026i,j,k,l,c,m]. None of those routes yielded a publication-safe positive mechanism claim.

2.3 Novelty Boundary and the Corrupted Watchlist

The repository’s novelty audit records an additional complication: early automated watchlists were corrupted by TeX-token overlap such as $\mathbf{P}^{n_1} \times \cdots \times \mathbf{P}^{n_s}$ and $\mathbb{Z}_2 \times \mathbb{Z}_2$, producing false positives in vector bundles, Lie theory, automorphic forms, and topology [Archivara Research Lab, 2026p,o]. Those papers are irrelevant to the arithmetic benchmark and are cited only as evidence that keyword silence is not novelty evidence. The real novelty burden comes from the arithmetic Kakeya line and from the danger of rebranding a failed direct-search regime as a successful local mechanism.

3 Background and Preliminaries

3.1 Arithmetic Kakeya and the Verifier-Side Task

Definition 1 (Arithmetic Kakeya exponent). Let $\alpha \geq 1$. For a finite set $G \subset \mathbb{Z}^2$ and a vector $\mathbf{x} = (x_1, x_2) \in \mathbb{Z}^2$, define

$$\mathbf{x} \cdot G = \{x_1 g_1 + x_2 g_2 : (g_1, g_2) \in G\}.$$

We say that $\text{AK}(\alpha)$ holds if, for every $\varepsilon > 0$, there exists a finite set $X \subset \mathbb{Z}^2$ such that $x_1 + x_2 \neq 0$ for every nonzero $\mathbf{x} \in X$ and

$$|(1, -1) \cdot G| \ll_\varepsilon \max_{\mathbf{x} \in X} |\mathbf{x} \cdot G|^{\alpha+\varepsilon}$$

for every finite $G \subset \mathbb{Z}^2$.

Conjecture 2 (Arithmetic Kakeya conjecture). *The endpoint statement $\text{AK}(1)$ holds.*

Theorem 3 (Geometric consequence of $\text{AK}(\alpha)$). *If $\text{AK}(\alpha)$ holds, then every Kakeya set $E \subset \mathbb{R}^d$ has Hausdorff dimension at least*

$$\alpha^{-1}d + 1 - \alpha^{-1}.$$

Theorem 3 is background motivation rather than a result of this paper; it follows from Bourgain’s slicing argument together with later progress that upgrades the Hausdorff-dimension conclusion [Bourgain, 1999, Leng et al., 2024].

The benchmark studied here uses the verifier-side certificate language supplied by the task.

Definition 4 (Verifier-side certificate). Fix a finite label set $X \subset \mathbb{Z}^2$ with $(0, 0) \in X$ and no nonzero label on the anti-diagonal line $\langle(1, -1)\rangle$. A verifier-side certificate over X consists of:

1. positive integers $d_1, \dots, d_k$ and a vertex set

$$V = [d_1] \times \dots \times [d_k], \quad [m] := \{1, \dots, m\};$$

2. edge-label functions

$$f_i : [d_1] \times \dots \times [d_{i-1}] \times [d_i - 1] \rightarrow X$$

for $1 \leq i \leq k$;

3. a set $T \subseteq V$ of initially solved vertices;
4. a finite set R of singleton-supported functions $r : V \rightarrow X$, meaning that each $r \in R$ is nonzero at exactly one vertex of V .

For a nonzero value $f_i(a_1, \dots, a_i) = x$, the verifier interprets this as an edge joining the two vertices whose first $i - 1$ coordinates equal $(a_1, \dots, a_{i-1})$ and whose i th coordinates are a_i and $a_i + 1$, with label x .

Definition 5 (Forcing and score). The verifier repeatedly applies the following operations:

1. add the signed two-vertex relation contributed by any graph edge;
2. if a relation is supported outside the current solved set on exactly one vertex u and has value $(a, -a)$ there with $a \neq 0$, then add u to the solved set;
3. add any integer linear combination of already available relations.

If all vertices are eventually solved, the certificate is *forcing*. Its score is

$$S(X, G, R, T) = \frac{m(G) + |R|}{n(G) - |T|},$$

where $n(G) = |V|$ and $m(G)$ counts the nonzero graph edges induced by the f_i .

The repo benchmark asks for a certificate with score at most 1.675 [Epoch AI, 2025, Archivara Research Lab, 2026n]. No such certificate appears in the saved artifacts.

3.2 Low-Height Grid Specialization

All saved search rows in the repository lie in a two-coordinate specialization of the verifier language.

Definition 6 (Grid instance). A *grid instance* is a tuple

$$C = (H, W, X, v, h, R, T)$$

with the following data.

1. integers $H \geq 2$ and $W \geq 1$;
2. a finite label set $X \subset \mathbb{Z}^2$ as above;
3. a vertical profile $v = (v_1, \dots, v_{H-1}) \in X^{H-1}$;
4. horizontal row data $h = (h_{r,c})_{1 \leq r \leq H, 1 \leq c \leq W-1}$ with each $h_{r,c} \in X$;
5. a finite set R of singleton seeds, each specified by a vertex $u \in [H] \times [W]$ and a nonzero label $x \in X$;
6. an initial solved set $T \subseteq [H] \times [W]$.

Definition 7 (Compiled grid certificate). Given a grid instance $C = (H, W, X, v, h, R, T)$, define $d_1 = H$ and $d_2 = W$. Set

$$f_1(i) = v_i \quad (1 \leq i \leq H-1),$$

and

$$f_2(r, c) = h_{r,c} \quad (1 \leq r \leq H, 1 \leq c \leq W-1).$$

Each singleton seed in R becomes a singleton-supported function on $[H] \times [W]$, and the set T is copied verbatim. The resulting verifier-side certificate is the *compiled* certificate associated to C .

This specialization allows different horizontal labels on different rows, but the vertical label between two adjacent rows is constant across columns. The Phase 4 corridor family is the special case $H = 2$.

3.3 Generator Vectors and Target Vectors

Fix a grid instance C with vertex set $V = [H] \times [W]$ and write $n = HW$. Enumerate V in row-major order:

$$(1, 1), (1, 2), \dots, (1, W), (2, 1), \dots, (H, W).$$

Each available verifier relation is identified with a vector in $\mathbb{Z}^{2n}$:

- a singleton seed at vertex u with label (a, b) contributes (a, b) in the two coordinates of u and zero elsewhere;
- an edge relation on vertices u and w with label (a, b) contributes (a, b) at u , $(-a, -b)$ at w , and zero elsewhere.

If $S \subseteq V$ is the current solved set, write $U = V \setminus S$ for the unsolved vertices and let M_U be the matrix whose columns are the current generator vectors projected to the coordinates indexed by U . For an unsolved vertex $u \in U$, let $e_u \in \mathbb{Q}^{2|U|}$ denote the anti-diagonal target vector supported only at u :

$$e_u = (0, \dots, 0, 1, -1, 0, \dots, 0).$$

Rule 2 can solve u precisely when a nonzero integer multiple of e_u lies in the integer column span of M_U .

Remark 8 (Scope). Every theorem proved below concerns the grid specialization from Definition 6. We do not claim a general sound-and-complete verifier theorem for arbitrary constructible graphs beyond this family.

4 Method

4.1 Evidence Boundary and Route Order

The paper uses only saved repository artifacts as experimental evidence. No new search row is run for the manuscript. The only new computations performed during writing are figure regeneration and LaTeX compilation. This boundary is deliberate: the verification artifacts explicitly recommend *REVISE* rather than a new experimental pass, provided the paper narrows every claim to the saved evidence [Archivara Research Lab, 2026q,b].

The repository proceeds in two stages. Phase 4 executed the corridor pilot routes H1 and H2. Phase 6 shifted to obstruction-first audits of typed residue interfaces, bounded shell libraries, passive-middle lifts, phase-coded schedules, controller search, and metastability summaries. Figure 1 shows the resulting pipeline: a single exact compiler and verifier serve both theorem-level obstructions and all benchmark rows.

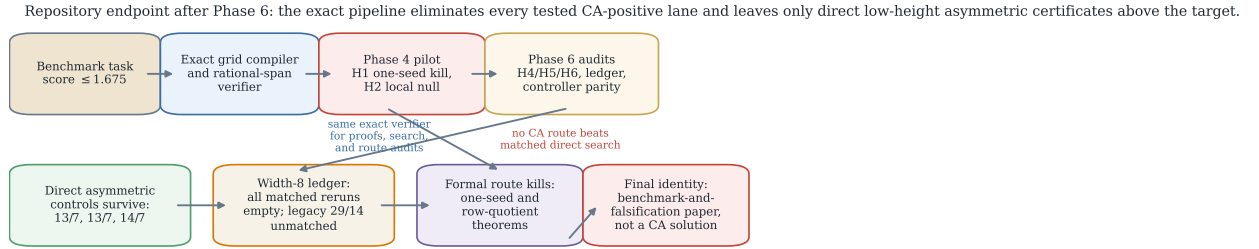


Figure 1: Repository pipeline at the current endpoint. The target is the exact verifier-side benchmark score 1.675. Every route, positive or negative, is tested through the same grid compiler and exact verifier. The Phase 4 corridor pilot kills H1 and gives a local null for H2; the Phase 6 audits eliminate typed interfaces, bounded shell grammars, passive-middle lifts, and defect-transport-style controller families. The only surviving positive objects are direct low-height asymmetric certificates at scores 13/7, 13/7, and 14/7. The figure is a workflow summary, not a performance plot.

4.2 Exact Verifier

The verifier used throughout the repo is exact. It never accepts a candidate because of rollout occupancy, closure size alone, or any approximate local statistic. At a given solved set S , it asks whether any anti-diagonal target e_u lies in the rational column span of the projected generator matrix M_U . If the answer is yes, the corresponding vertex may be solved. Otherwise the certificate fails.

Algorithm 1: Exact verifier for a compiled grid certificate

1. Input a compiled grid certificate and initialize the solved set $S \leftarrow T$.
2. While $S \neq V$:
 - (a) set $U \leftarrow V \setminus S$ and build the projected matrix M_U ;
 - (b) for each $u \in U$, test whether $e_u \in \text{span}_{\mathbb{Q}}(\text{columns}(M_U))$;
 - (c) if no such u exists, return failure;
 - (d) otherwise solve one such u and continue.
3. Return the forcing order.

The exact implementation uses rational linear algebra for paper-facing decisions. The repo also contains fast closure heuristics, based on floating-point rank tests or modular filters, but those are used only inside proposal generation. Any exact score, exact hit rate, exact frontier row, or exact forcing claim in this paper is backed by Algorithm 1 rather than by a surrogate [Archivara Research Lab, 2026b,k].

4.3 Greedy Seed Search and Direct Baselines

Most positive rows in the repository come from fixed geometry plus a direct search over seeds and, in some phases, over edge labels. For a fixed choice of H , W , X , vertical profile v , horizontal labels h , seed budget, initial- T budget, and boundary band, the repo uses a greedy proposal rule.

Algorithm 2: Greedy seed search on a fixed grid family

1. Enumerate candidate boundary vertices according to the boundary band.
2. For each allowed initial solved set T_0 of the prescribed size:
 - (a) initialize the seed set $R_0 \leftarrow \emptyset$;
 - (b) if the current certificate is already forcing, return it;
 - (c) for each remaining seed slot:
 - i. evaluate every admissible new singleton seed using a fast closure oracle;
 - ii. keep the seed that maximizes closure size;
 - iii. if exact verification now succeeds, return the exact forcing certificate.
3. If no forcing certificate is found, return failure.

Random direct search wraps Algorithm 2 in an outer loop that samples the edge labels themselves. Deterministic reruns fix the edge pattern and use Algorithm 2 only for seeds and initial solved vertices. In either case, a candidate is recorded only if Algorithm 1 certifies forcing.

4.4 Route-Specific Audit Protocols

Phase 4 corridor pilot. The Phase 4 corridor pilot froze a same-sum width-2 family. The H1 representative used a single singleton seed, empty initial solved set, one nonzero top-row label, zero

bottom row, and a constant vertical label. H2 then computed an abelian-style invariant package on the four family IDs that mattered to the pilot: H1 level 1, H1 level 2, the width-4 direct control, and the width-6 direct control [Archivara Research Lab, 2026h,g,i]. These rows remain useful because they establish the initial corridor obstruction and a local null result for the imported H2 feature package.

H4 typed residue interfaces. The Phase 6 champion route fixes one low-height asymmetric family

$$X = \{(0, 0), (1, 0), (0, 1), (1, 1), (2, -1)\},$$

one interface alphabet

$$\{\text{closed}, \text{top}, \text{bottom}, \text{gate_up}, \text{gate_down}\},$$

and one exact extractor. A canonicalization quotient identifies words up to horizontal reversal and the row-swap involution $\text{top} \leftrightarrow \text{bottom}$, $\text{gate_up} \leftrightarrow \text{gate_down}$, leaving 39 canonical width-4 interface words of length 3. Each word is compiled to an $H = 2$, $W = 4$ grid, passed through Algorithm 2 with seed budget 4, initial- T budget 1, and boundary band 1, and then checked by Algorithm 1. Level-2 composition is allowed only if level-1 forcing words exist; composition must use the same alphabet, the same extractor, and no repair pass [Archivara Research Lab, 2026j].

Complexity-accounted width-8 ledger. The width-8 ledger re-runs one deterministic pattern for each of four family types: same-palette, low-height asymmetric, bounded-slope, and slowly growing X . Every row uses the same geometry ($H = 2$, $W = 8$), the same seed budget 3, initial- T budget 1, and boundary band 1. The ledger records not only score but also $|X|$, coordinate height, grammar length, active state count, extractor complexity, and boundary initialization. The saved exploratory 29/14 witness is retained only as a legacy row outside that matched ledger [Archivara Research Lab, 2026d].

H5 affine-core / nonlinear-shell audit. This audit keeps the H4 extractor fixed and splits the interface alphabet into an affine core

$$\{\text{closed}, \text{top}, \text{bottom}\}$$

and a nonlinear shell

$$\{\text{gate_up}, \text{gate_down}\}.$$

For each of the three low-height asymmetric families asym_a , asym_b , and asym_c , the repo checks all 39 canonical shell words on a width-4 grid with seed budget 4, initial- T budget 1, and boundary band 1. The matched direct comparator runs 20 exact trials at each of two seeds for the same family and budgets [Archivara Research Lab, 2026k].

Passive-middle geometry lift. The height-3 lift keeps the H4 extractor but inserts a passive middle row. Horizontal labels are allowed only on the top and bottom rows; the middle row has no horizontal edges. Both vertical layers use the same label c . Seeds and initial solved vertices are restricted to the top and bottom boundary rows. The audit checks all 39 canonical width-4 route words for each low-height asymmetric family and compares them to unrestricted direct search with 10 exact trials at each of two seeds [Archivara Research Lab, 2026e].

H6 defect transport, controller parity, and metastability. The H6 family fixes one asymmetric label set and one width-4 motif alphabet $\{S, D, P\}$. The periodic word is SDS, the aperiodic word is SDP, and the reversed word is PDS. The route is tested at $H = 2$ and $H = 3$, with seed budget 3, initial- T budget 1, and a thinned-seed robustness check at budget 2 [Archivara Research Lab, 2026l]. A separate exhaustive controller solver enumerates all 75,276 programs in the frozen controller bank and compares the best controller certificate to the matched direct archive [Archivara Research Lab, 2026c]. Finally, the metastability audit scans all 27 words in the same bank and records the best closure-time program for each word, together with exact score and a dependency-hypergraph summary [Archivara Research Lab, 2026m].

4.5 Complexity and Search-Space Accounting

Let $n = HW$ be the number of vertices and let $g = m + r$ be the number of initial generators in a compiled grid instance. At one verifier stage with u unsolved vertices, the projected matrix M_U has $2u$ rows and g columns. Dense exact elimination on M_U has polynomial cost in u , g , and coefficient size; a coarse dense bound is $O(u^2g + u^3)$ arithmetic operations per span test. Testing all vertices across all stages therefore yields a worst-case bound polynomial in n and g , with the saved repo rows remaining tiny enough that exact auditability is preferable to asymptotic scale.

The route audits are finite and explicit. Table 1 records the exact search spaces, trial budgets, and seeds. The critical point is methodological rather than asymptotic: the repo freezes and reports the proposal bank itself. A route cannot claim lift by silently enlarging X , adding interface symbols, changing the extractor, or loosening the boundary budget [Archivara Research Lab, 2026b,d].

5 Theoretical Results and Proofs

5.1 Compiler and Verifier Correctness

Proposition 9. *Let $C = (H, W, X, v, h, R, T)$ be a grid instance. Its compiled object from Definition 7 is a legal verifier-side certificate. The compiled graph has*

$$n = HW$$

vertices and

$$m = W \sum_{i=1}^{H-1} 1_{v_i \neq (0,0)} + \sum_{r=1}^H \sum_{c=1}^{W-1} 1_{h_{r,c} \neq (0,0)}$$

nonzero edges.

Proof. The compiled certificate has $d_1 = H$ and $d_2 = W$, so its vertex set is $[H] \times [W]$ and therefore has size HW .

We next check legality of the edge-label functions. For f_1 , the domain is $[H - 1]$, and $f_1(i) = v_i \in X$ by definition of a grid instance. For f_2 , the domain is $[H] \times [W - 1]$, and $f_2(r, c) = h_{r,c} \in X$. Hence every nonzero edge label used by the compiled certificate belongs to $X \setminus \{(0, 0)\}$, as required.

Now consider the singleton seeds. Every element of R is, by definition of a grid instance, supported on a single vertex $u \in [H] \times [W]$ and takes a nonzero value in X . Therefore each compiled seed is a legal singleton-supported verifier function. The initial solved set T is a subset of the vertex set, so it is also legal.

It remains to count nonzero edges. For each inter-row layer $i \in \{1, \dots, H-1\}$ and each column $c \in \{1, \dots, W\}$, the label v_i contributes one vertical edge exactly when $v_i \neq (0, 0)$. Summing over columns gives the term

$$W \sum_{i=1}^{H-1} 1_{v_i \neq (0,0)}.$$

For horizontal edges, each pair (r, c) with $1 \leq r \leq H$ and $1 \leq c \leq W-1$ contributes one nonzero edge exactly when $h_{r,c} \neq (0, 0)$. Summing these indicators gives

$$\sum_{r=1}^H \sum_{c=1}^{W-1} 1_{h_{r,c} \neq (0,0)}.$$

Adding the vertical and horizontal contributions yields the stated formula for m . $\square$

Lemma 10. *Let M be an integer matrix and let e be a rational vector. The following are equivalent:*

1. *there exists a nonzero integer a such that $ae \in \text{span}_{\mathbb{Z}}(\text{columns}(M))$;*
2. *$e \in \text{span}_{\mathbb{Q}}(\text{columns}(M))$.*

Proof. Assume (1). Then $ae = Mz$ for some integer vector z . Dividing by the nonzero integer a gives

$$e = M \left(\frac{1}{a} z \right),$$

so e lies in the rational column span of M . Thus (1) $\Rightarrow$ (2).

Assume (2). Then $e = Mq$ for some rational vector q . Let $D > 0$ be a common denominator of the coordinates of q . Then Dq is an integer vector, so

$$De = M(Dq) \in \text{span}_{\mathbb{Z}}(\text{columns}(M)).$$

Because D is a nonzero integer, (1) holds with $a = D$. Thus (2) $\Rightarrow$ (1). $\square$

Lemma 11. *Fix a compiled grid certificate and a solved set S . Let $U = V \setminus S$ and let M_U be the projected generator matrix. For an unsolved vertex $u \in U$, the following are equivalent:*

1. *the verifier may solve u by one application of rule 2 at this stage;*
2. *$e_u \in \text{span}_{\mathbb{Q}}(\text{columns}(M_U))$.*

Proof. The verifier may solve u exactly when some available relation is supported outside S on the single vertex u and has value $(a, -a)$ there for some nonzero integer a . In projected coordinates this means precisely that

$$ae_u \in \text{span}_{\mathbb{Z}}(\text{columns}(M_U))$$

for some nonzero integer a . By Lemma 10, this is equivalent to

$$e_u \in \text{span}_{\mathbb{Q}}(\text{columns}(M_U)).$$

$\square$

Theorem 12. *For the grid specialization from Definition 6, Algorithm 1 is sound and complete: it returns a full forcing order if and only if the compiled grid certificate is forcing.*

Proof. We first prove soundness. Suppose Algorithm 1 returns a full order of the vertices. At each step, the algorithm adds a vertex u only after checking that e_u lies in the rational column span of the current projected generator matrix. By Lemma 11, that condition is equivalent to the existence of a legal rule 2 witness for u at that stage. Therefore every step taken by the algorithm is legal in the verifier semantics. Because the algorithm eventually solves every vertex, the certificate is forcing.

We next prove completeness. Assume that the compiled certificate is forcing. We induct on the number of unsolved vertices.

If all vertices are already solved, the claim is immediate. Assume now that at least one vertex is unsolved and that the theorem has already been proved for every forcing instance with fewer unsolved vertices.

Because the certificate is forcing, there exists at least one legal next rule 2 step from the current solved set S . Choose a vertex u that can be solved in such a step. By Lemma 11, e_u lies in the rational column span of the current projected generator matrix, so Algorithm 1 will include u among its admissible candidates.

Algorithm 1 may choose u or it may choose another admissible candidate u' . We claim that solving one admissible vertex cannot destroy the admissibility of another admissible vertex. Indeed, suppose u' is admissible from S . Then there exists a relation whose support outside S is exactly $\{u'\}$ and whose value there is anti-diagonal. If we enlarge the solved set from S to $S \cup \{u\}$ with $u \neq u'$, the same relation still has support outside the larger solved set equal to $\{u'\}$, because every coordinate that was allowed to be nonzero on the solved set remains allowed and one additional coordinate has become solved. Therefore u' remains admissible after solving u .

Now take any full forcing sequence from S . If its first solved vertex is already the vertex chosen by Algorithm 1, remove that step and apply the induction hypothesis to the remaining suffix. Otherwise, if the first vertex in that forcing sequence is some other admissible vertex u' , the previous paragraph shows that Algorithm 1's chosen vertex is also admissible at the same stage and that solving it first does not destroy the admissibility of u' . Hence we may swap Algorithm 1's chosen vertex to the front of the forcing sequence. After that swap, the remaining unsolved problem has one fewer vertex than before and is still forcing. By the induction hypothesis, Algorithm 1 solves the remainder.

Thus Algorithm 1 returns a full forcing order whenever the compiled certificate is forcing. Together with soundness, this proves the theorem. $\square$

5.2 Obstruction Theorems

Theorem 13 (One-seed obstruction). *Let $X \subset \mathbb{Z}^2$ be finite, and assume that no nonzero element of X lies in $\langle (1, -1) \rangle$. Consider any legal verifier-side certificate with empty initial solved set and exactly one initial singleton seed,*

$$R = \{xu_0\},$$

where u_0 is a vertex and $x \in X \setminus \{(0, 0)\}$. Then the certificate is not forcing.

Proof. Let Σ denote the total-label map that sums a relation over all vertices and returns an element of $\mathbb{Z}^2$. We inspect the three verifier operations.

Step 1: edge generators have zero total label. An edge generator with label y places y at one endpoint and $-y$ at the other endpoint, so its total label is

$$y + (-y) = 0.$$

Step 2: the unique initial seed has total label x . This is immediate because the seed is supported at a single vertex.

Step 3: every generated relation has total label in $\mathbb{Z}x$. Initially, the available nonzero relations are the unique seed, of total label x , and the edge generators, all of total label 0. Thus the claim holds at time zero. If two already available relations have total labels k_1x and k_2x , then any integer linear combination of them has total label $(ak_1 + bk_2)x$ for integers a, b . Therefore the claim is preserved by rule 3, and it is obviously preserved by adding more edge generators of total label 0.

Now suppose, for contradiction, that the certificate is forcing. Because the initial solved set is empty, some first rule 2 step must eventually occur. At that first rule 2 step, there exists a generated relation whose support is a single vertex and whose value there is $(a, -a)$ with $a \neq 0$. Since no vertex has yet been solved, the total label of that relation is exactly $(a, -a)$. By Step 3, this total label must lie in $\mathbb{Z}x$, so there exists $k \in \mathbb{Z}$ such that

$$kx = (a, -a).$$

If $k = 0$, then $(a, -a) = (0, 0)$, contradicting $a \neq 0$. If $k \neq 0$, then

$$x = \frac{a}{k}(1, -1),$$

so $x \in \langle (1, -1) \rangle$, contradicting the hypothesis on X .

Therefore no first rule 2 step exists. The solved set can never grow from the empty set, so the certificate is not forcing. $\square$

Corollary 14. *Every frozen H1 pilot instance in the repository fails exact verification before scoring.*

Proof. The saved obstruction artifact records that every frozen H1 level has exactly one singleton seed and empty initial solved set [Archivara Research Lab, 2026g]. The active same-sum palette contains no nonzero anti-diagonal label, so Theorem 13 applies directly. $\square$

Theorem 15 (Passive-middle row-quotient obstruction). *Fix a nonzero vector $c \in \mathbb{Z}^2$ with $c \notin \langle (1, -1) \rangle$. Let $W \geq 1$, and consider a height-3 grid instance with the following properties:*

1. *the vertical profile is (c, c) ;*
2. *every horizontal label on the middle row is $(0, 0)$;*
3. *every singleton seed and every initially solved vertex lies on the top or bottom row.*

Then the compiled certificate is not forcing.

Proof. For each column $j \in \{1, \dots, W\}$, let $m_j = (2, j)$ be the middle-row vertex in that column, and let π_j be the projection map that extracts the two coordinates of a relation at the vertex m_j .

We first prove that every available relation has middle-row projection in the rank-one lattice $\mathbb{Z}c$ at each column.

Singleton seeds. By hypothesis, singleton seeds are supported only on the top or bottom row. Therefore π_j sends every singleton seed to 0, which lies in $\mathbb{Z}c$.

Horizontal edge generators. A horizontal edge on the top or bottom row does not touch m_j , so its π_j -projection is $0 \in \mathbb{Z}c$. A horizontal edge on the middle row has label $(0, 0)$ by assumption, so its generator is the zero relation and again has π_j -projection $0 \in \mathbb{Z}c$.

Vertical edge generators. At column j , the upper vertical edge contributes either c or $-c$ at m_j , and the lower vertical edge contributes either c or $-c$ at m_j . Hence the π_j -projection of every nonzero vertical edge generator lies in $\mathbb{Z}c$.

Because rule 1 only adds edge generators of the three types above and rule 3 only takes integer linear combinations, it follows that every relation ever generated satisfies

$$\pi_j(r) \in \mathbb{Z}c$$

for every column j .

Now suppose, toward contradiction, that the certificate is forcing. Since no middle-row vertex lies in the initial solved set, there must be a first time at which some middle-row vertex is solved by rule 2. Let that vertex be m_j . At that moment, rule 2 provides a relation r whose support outside the already solved vertices is exactly $\{m_j\}$ and whose value at m_j is $(a, -a)$ with $a \neq 0$. In particular,

$$\pi_j(r) = (a, -a).$$

But the previous paragraph shows that $\pi_j(r) \in \mathbb{Z}c$, so there exists $k \in \mathbb{Z}$ with

$$kc = (a, -a).$$

If $k = 0$, then $(a, -a) = (0, 0)$, contradicting $a \neq 0$. If $k \neq 0$, then

$$c = \frac{a}{k}(1, -1),$$

which means that $c \in \langle(1, -1)\rangle$, contradicting the hypothesis.

Therefore no middle-row vertex can ever be solved. Since the initial solved set contains no middle-row vertex and a forcing certificate must eventually solve every vertex, the compiled certificate is not forcing. $\square$

Corollary 16. *The passive-middle $H = 3$ lift audited in the repository is impossible on the families `asym_a`, `asym_b`, and `asym_c`.*

Proof. In the saved geometry-lift audit, the three vertical labels are $c = (1, 1)$, $c = (2, -1)$, and $c = (1, 2)$, whose coordinate sums are 2, 1, and 3 respectively [Archivara Research Lab, 2026e]. Hence none lies in $\langle(1, -1)\rangle$. The audit also fixes zero horizontal labels on the middle row and restricts seeds and initial solved vertices to the top and bottom boundary rows. Therefore Theorem 15 applies to each family. $\square$

6 Experimental Setup

6.1 Saved Artifacts Used as Evidence

Table 1 lists the saved artifacts used in the paper. The table is intentionally explicit: the benchmark contribution here is as much about auditable route accounting as it is about any single score.

Phase	Artifact	Geometry / family	Search space or budget	Role in the paper
4	H1 obstruction	width-2 corridor, same-sum palette	exact symbolic route invariant, no RNG	theorem-backed kill of the frozen macrocell pilot
4	H2 screen	four family IDs	fixed feature computation, no RNG	local null result for the abelian-style descriptor package
6	H4 typed interfaces	asym_a, $H = 2$, $W = 4 \rightarrow 8$	39 canonical width-4 words; matched direct width-8 comparator with 60 exact trials	interface-sufficiency audit
6	width-8 complexity ledger	four matched width-8 families plus one legacy row	four deterministic reruns at seed budget 3, initial- T budget 1, boundary band 1	hidden-complexity accounting for the width-8 story
6	H5 shell audit	asym_a, asym_b, asym_c, $H = 2$, $W = 4$	$3 \times 39 = 117$ canonical shell words; matched direct comparator = $3 \times 40 = 120$ exact trials	bounded-shell mechanism audit and direct frontier extraction
6	geometry lift	asym_a, asym_b, asym_c, $H = 3$, $W = 4$	117 route words; matched unrestricted direct comparator = 60 exact trials	passive-middle row-quotient audit
6	H6 schedule tests	asym_a, $H = 2$ and $H = 3$	five fixed schedule rows, including periodic, aperiodic, thinned, reversed, and lifted variants	defect-transport falsification
6	controller enumeration	asym_a, $H = 2$, $W = 4$	exhaustive bank of 75,276 programs	parity check against the direct frontier
6	metastability scan	asym_a, $H = 2$, $W = 4$	all 27 words in the frozen motif bank, each with exhaustive boundary seeds / initial- T choices	test whether closure-time summaries predict score

Table 1: Saved artifacts used as evidence. Every route-facing score or hit rate in the paper is backed by the exact verifier, not by rollout proxies [Archivara Research Lab, 2026g,i,j,d,k,e,l,c,m].

6.2 Budgets, Random Seeds, and Exhaustive Enumerations

The saved budgets are part of the evidence:

- H4 uses seed budget 4, initial- T budget 1, boundary band 1, and two direct-search seeds (3101,3102) with 30 exact trials each [Archivara Research Lab, 2026j].

- The width-8 complexity ledger fixes seed budget 3, initial- T budget 1, and boundary band 1 for all matched reruns [Archivara Research Lab, 2026d].
- H5 uses seed budget 4, initial- T budget 1, boundary band 1, and direct-search seeds (6201, 6202) with 20 exact trials each per family [Archivara Research Lab, 2026k].
- The geometry-lift direct comparator uses seeds (7201, 7202) with 10 exact trials each per family [Archivara Research Lab, 2026e].
- H6 fixed schedules use seed budget 3, initial- T budget 1, boundary band 1, together with one thinned-seed test at budget 2 [Archivara Research Lab, 2026l].
- The controller bank and metastability scan are exhaustive over their frozen finite banks and therefore use no randomness [Archivara Research Lab, 2026c,m].

6.3 Reproducibility Caveats

The repository saves command lines, seeds, geometry, budgets, and exact JSON outputs, but it does not preserve publication-grade hardware logs for the original search runs. We therefore do *not* make runtime or throughput claims. The reproducibility burden carried by this paper is exact logical and exact numerical reproducibility: same scripts, same seeds, same route banks, same verifier, same saved outputs.

7 Results

Table 2 gives the route-level summary before we discuss the routes one by one.

Route	Exact search space	Route outcome	Matched direct comparator	Evidence-based conclusion
H1	theorem plus frozen pilot instances	no forcing row at any level	phase-4 corridor controls at score 2.0	killed by Theorem 13; no score comparison is meaningful
H2	four saved family IDs	no ranking lift beyond raw support and target-solvable counts	none needed beyond the same four IDs	local null result only
H4	39 canonical width-4 interface words	0 forcing words, hence 0 compositions	matched width-8 direct hit rate 0/60	typed interfaces fail before composition
ledger	four matched width-8 reruns plus one legacy row	all matched rows empty; only legacy 29/14 row scores	same matched reruns	the saved 29/14 witness is outside the matched envelope
H5	117 canonical shell words across three families	0 forcing shell words	direct search finds 13/7, 13/7, and 14/7	bounded shell library loses to direct search
$H = 3$ lift	117 passive-middle route words plus Theorem 15	no forcing route word on any family	direct hit rate 0/20 on each family	passive-middle lift is killed by a row quotient
H6	periodic / aperiodic / thinned / reversed / lifted schedules	only SDP at $H = 2$ forces, with score 13/7	the same 13/7 direct certificate already exists	no mechanism lift survives robustness checks
controller	75,276 exact programs	48 forcing programs; best word SDP at 13/7	best direct certificate also 13/7	exhaustive controller search recovers parity, not improvement
metastability	27 words with exhaustive boundary programs	four-word tie at score 13/7; many high-closure failures remain nonforcing	direct representative SDP belongs to the tie	closure-time summaries do not predict better certificates

Table 2: Route summary at the repository endpoint. Every number in the table is an exact theorem statement or an exact measurement backed by a saved artifact [Archivara Research Lab, 2026g,i,j,d,k,e,l,c,m].

7.1 The Phase 4 Pilot Dies by Proof, and H2 Adds Only a Local Null Result

The frozen H1 pilot is a theorem-level failure, not a poor score. Corollary 14 shows that every saved H1 level has one singleton seed and empty initial solved set, so exact verification fails before scoring. Figure 2 reminds the reader what that family looks like, while Figure 3a translates the proof into the invariant picture used by the repo.

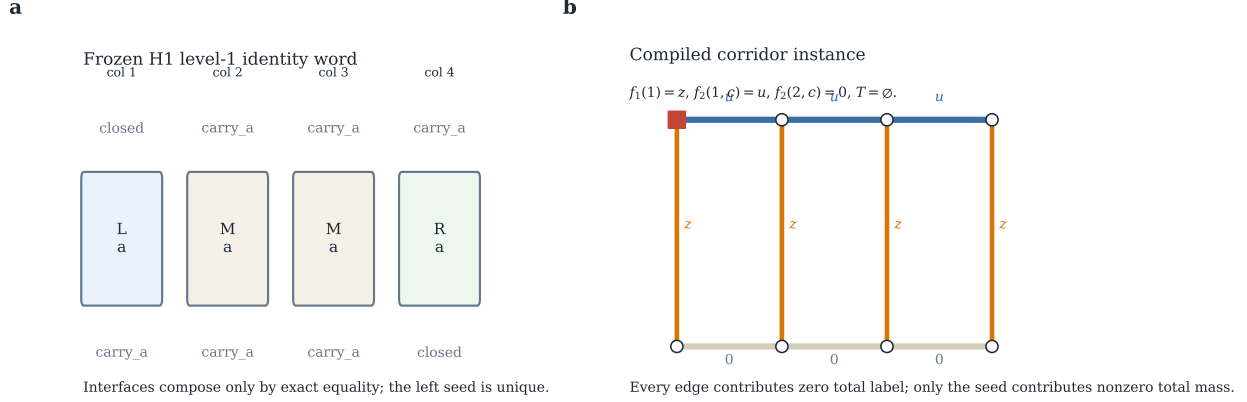


Figure 2: Frozen H1 corridor pilot. **a**, the saved level-1 identity word. **b**, its exact compiled corridor instance: one singleton seed at the left boundary, empty initial solved set, one nonzero top-row label, zero bottom row, and constant vertical label. This is precisely the setting of Corollary 14. The figure remains relevant even after Phase 6 because it visualizes the pilot route that first failed by proof.

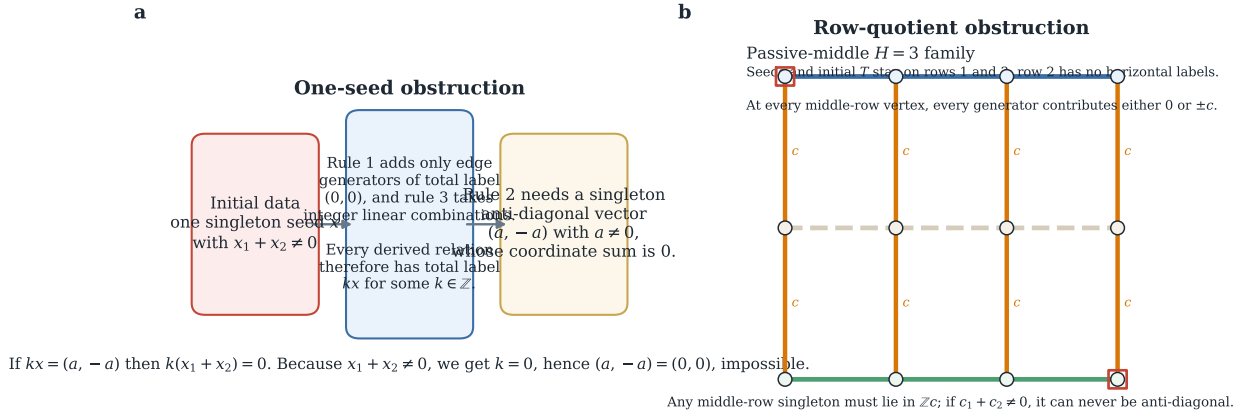


Figure 3: The two formal obstructions proved in Section 5. **a**, the one-seed obstruction: every generated relation has total label in the rank-one lattice generated by the unique seed, but rule 2 requires a nonzero anti-diagonal singleton. **b**, the passive-middle row-quotient obstruction: in the height-3 lift, every middle-row contribution is a multiple of the vertical label c , while the middle row must eventually produce an anti-diagonal singleton to force. The first panel explains the Phase 4 H1 failure; the second anticipates the Phase 6 geometry-lift failure.

The H2 feature package is weaker still. It is not a failed theorem route, but a local screening null on a four-family set. Table 3 and Figure 7 show that raw support size and target-solvable count

already separate the dead H1 rows from the surviving direct controls. The imported abelian-style descriptors do not improve the ranking on this tiny screened set [Archivara Research Lab, 2026i].

Family	forcing?	r	rank	Smith tail	m/n	target-solvable
H1 level 1	no	1	8	1	0.875	0
H1 level 2	no	1	16	1	0.9375	0
phase-4 width 4 control	yes	5	12	2	0.875	2
phase-4 width 6 control	yes	7	20	2	1.0833	1

Table 3: The saved H2 feature screen. The table is intentionally narrow: it covers only the four family IDs actually screened in the repository, and no claim beyond that scope is warranted [Archivara Research Lab, 2026i,q].

7.2 The Best Saved Exact Certificates Are Direct Low-Height Asymmetric Controls

Phase 6 changes the repo-wide frontier. Table 4 and Figure 4a show that the best currently saved exact certificates are no longer the phase-4 corridor controls at score 2.0, but the low-height asymmetric direct certificates at score $13/7$ on `asym_a` and `asym_b` [Archivara Research Lab, 2026k]. The best `asym_c` certificate is $14/7 = 2.0$, tying rather than improving the phase-4 width-4 and width-6 controls.

Family	Status	Geometry	Exact score	Note
phase-4 width-4 control	matched direct baseline	$H = 2,$ $W = 4$	2.0	corridor pilot
phase-4 width-6 control	matched direct baseline	$H = 2,$ $W = 6$	2.0	corridor pilot
<code>asym_a</code>	matched direct baseline	$H = 2,$ $W = 4$	$13/7 \approx 1.8571$	best current exact row
<code>asym_b</code>	matched direct baseline	$H = 2,$ $W = 4$	$13/7 \approx 1.8571$	best current exact row
<code>asym_c</code>	matched direct baseline	$H = 2,$ $W = 4$	$14/7 = 2.0$	ties the old corridor controls
legacy width-8 witness	exploratory unmatched row	$H = 2,$ $W = 8$	$29/14 \approx 2.0714$	excluded by the matched ledger

Table 4: Positive exact rows saved in the repository. The low-height asymmetric certificates from Phase 6 dominate the Phase 4 corridor pilot. The legacy width-8 row is retained for transparency but is not part of the matched benchmark frontier [Archivara Research Lab, 2026f,r,s,k,d].

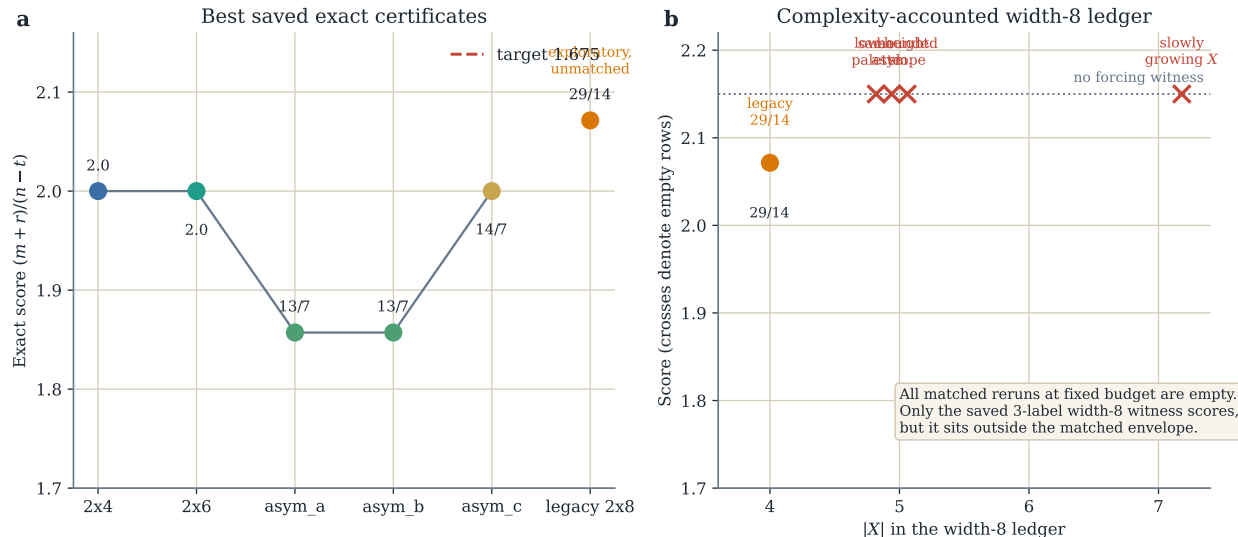


Figure 4: Direct frontier and width-8 ledger. **a**, the best saved exact positive certificates: the Phase 4 corridor controls at score 2.0, the Phase 6 low-height asymmetric controls at 13/7, 13/7, and 14/7, and the unmatched legacy width-8 witness at 29/14. The red dashed line marks the target 1.675. **b**, the complexity-accounted width-8 ledger. Every matched rerun at fixed budget is empty; the only score-bearing row is the saved exploratory 29/14 witness, which uses a different palette and therefore sits outside the matched envelope.

7.3 Typed Interfaces Fail Before Composition, and the Width-8 Ledger Stays Empty

The H4 interface audit is the cleanest saved test of whether a finite cellular-automaton interface alphabet carries real arithmetic content. It does not. The route checks 39 canonical width-4 words and finds zero forcing words, so the level-2 composition space is empty by definition. The matched direct width-8 search on the same family and budgets also finds no witness in 60 exact trials [Archivara Research Lab, 2026j]. Figure 6a reports the H4 hit rate together with its Wilson interval.

The complexity-accounted width-8 ledger sharpens the same negative story. All four matched reruns—same palette, low-height asymmetric, bounded slope, and slowly growing X —remain empty. The only score-bearing width-8 row is the saved exploratory three-label witness at 29/14. Figure 4b therefore matters as much as Figure 4a: score-only ranking would suggest width-8 progress, but the matched ledger says otherwise [Archivara Research Lab, 2026d].

7.4 The Bounded Shell Audit Loses to Direct Search

The H5 audit turns the local-mechanism question into a finite competition: can a frozen two-gate shell library beat either the affine core or unrestricted direct search under the same extractor and the same budgets? Across the three low-height asymmetric families, the answer is no. The shell audit checks 39 canonical words per family, hence 117 in total, and finds zero forcing shell words. The matched direct comparator finds forcing certificates on all three families, with hit rate 0.2 over 40 exact trials in each case and best scores 13/7, 13/7, and 14/7 [Archivara Research Lab, 2026k].

Figure 6a aggregates the shell and direct hit rates. Figure 5 renders two representative direct

certificates. Panel (a) is the best `asym_a` row at score 13/7; panel (b) shows the best `asym_c` row at score 14/7. The direct frontier is visibly boundary-loaded, but it is exact and legal, which the shell route is not.

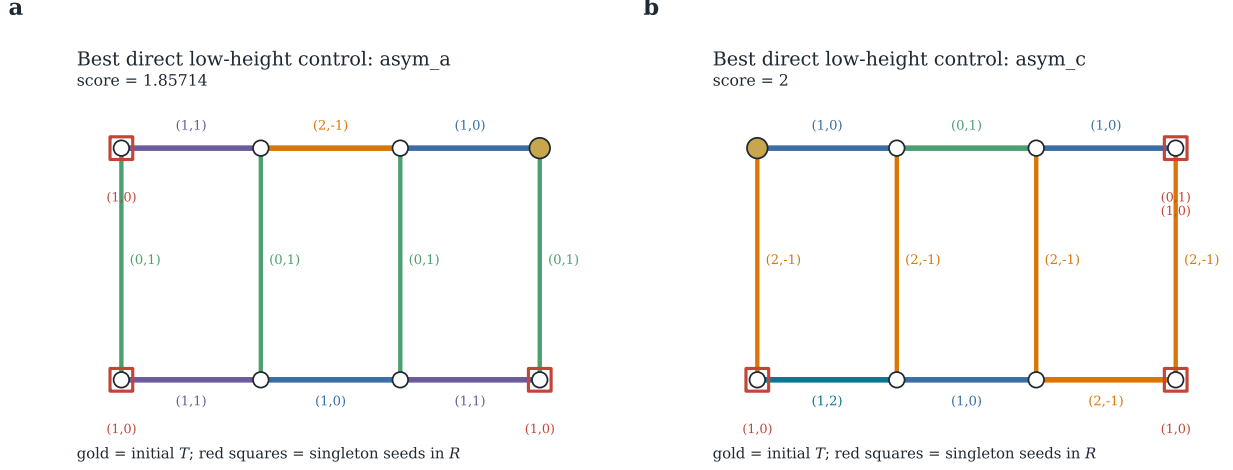


Figure 5: Best saved direct low-height asymmetric certificates. **a**, the `asym_a` certificate at score 13/7; the `asym_b` frontier row has the same score with a different label set. **b**, the `asym_c` certificate at score 14/7. Gold vertices belong to the initial solved set, red squares mark singleton seeds, and colored edges indicate nonzero labels. These are the best exact positive rows currently saved in the repository [Archivara Research Lab, 2026k].

7.5 The Passive-Middle Geometry Lift Is Killed by a Row Quotient

Theorem 15 predicts that the passive-middle height-3 lift cannot force, regardless of width, as long as the middle row stays horizontal-zero and the vertical label c is not anti-diagonal. The Phase 6 geometry-lift artifact confirms that prediction on all three low-height asymmetric families: each family checks 39 route words and finds zero forcing words, while the matched unrestricted direct comparator also records hit rate 0.0000 over 20 exact trials [Archivara Research Lab, 2026e]. Figure 3b is therefore both a proof diagram and a faithful picture of the saved route family.

This obstruction is formally stronger than the one-seed theorem. The one-seed argument rules out a specific seed pattern. The row-quotient argument rules out an entire higher-geometry family while allowing arbitrary boundary seeds and arbitrary width inside that family. In the current repository, it is the strongest theorem-level negative contribution.

7.6 H6 Schedules, Exhaustive Controllers, and Metastability Show No Lift

The H6 schedule family produces exactly one apparent positive row: the aperiodic word SDP at height 2 forces with score 13/7. But the saved defect-transport audit immediately removes the mechanism-level interpretation. The periodic word SDS does not force, the same SDP word fails under seed thinning and reversal, and the height-3 lift does not force at all [Archivara Research Lab, 2026l]. Moreover, the saved direct archive already contains the exact same 13/7 certificate.

The controller audit reinforces the same point. Exhaustive enumeration of 75,276 exact controller

programs yields 48 forcing programs and identifies SDP as the best word, again with score 13/7 [Archivara Research Lab, 2026c]. This is not a new frontier. It is parity with the matched direct row.

Finally, the metastability scan shows that closure-time and dependency-hypergraph summaries do not identify any better certificate than the direct frontier. Among the 27 words in the frozen motif bank, the equal-best score 13/7 is achieved by four words,

$$\{\text{DPS}, \text{PDS}, \text{SDP}, \text{SPD}\},$$

and the direct representative SDP is already among them. Many high-closure failures remain nonforcing [Archivara Research Lab, 2026m]. Figure 6 summarizes all three facts.

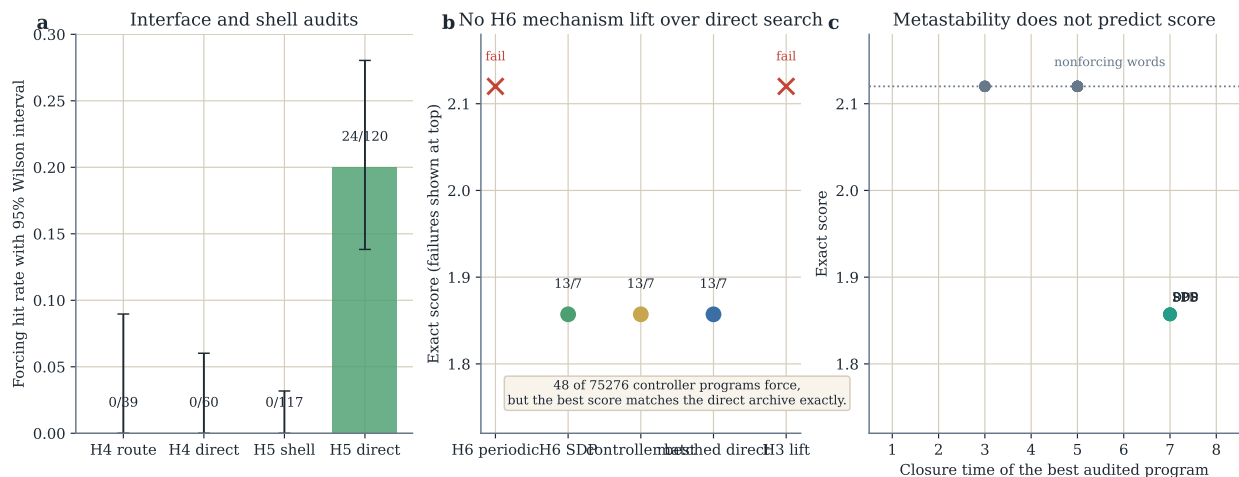


Figure 6: Phase 6 route audits beyond the corridor pilot. **a**, forcing hit rates with 95% Wilson intervals: H4 typed interfaces and H5 shell words both stay at zero, while the aggregated H5 direct comparator remains positive. **b**, H6 parity: the only positive schedule is the aperiodic word SDP at score 13/7, and the exhaustive controller bank recovers the same best score rather than a stronger one. **c**, metastability scatter: high closure time does not predict score, and the matched direct representative SDP lies inside a four-word tie rather than below it.

7.7 The H2 Feature Screen Remains a Local Null

The H2 screen is no longer the most important result in the repo, but it remains instructive. Figure 7 shows exactly what the Phase 4 screen established and, equally importantly, what it did not establish. On the four screened family IDs, raw support size and target-solvable count already separate the dead H1 rows from the surviving direct controls. The imported H2 descriptors do not improve the decision. This remains a local null on a tiny family set, not a theorem about abelian-style screening more generally [Archivara Research Lab, 2026i,q].

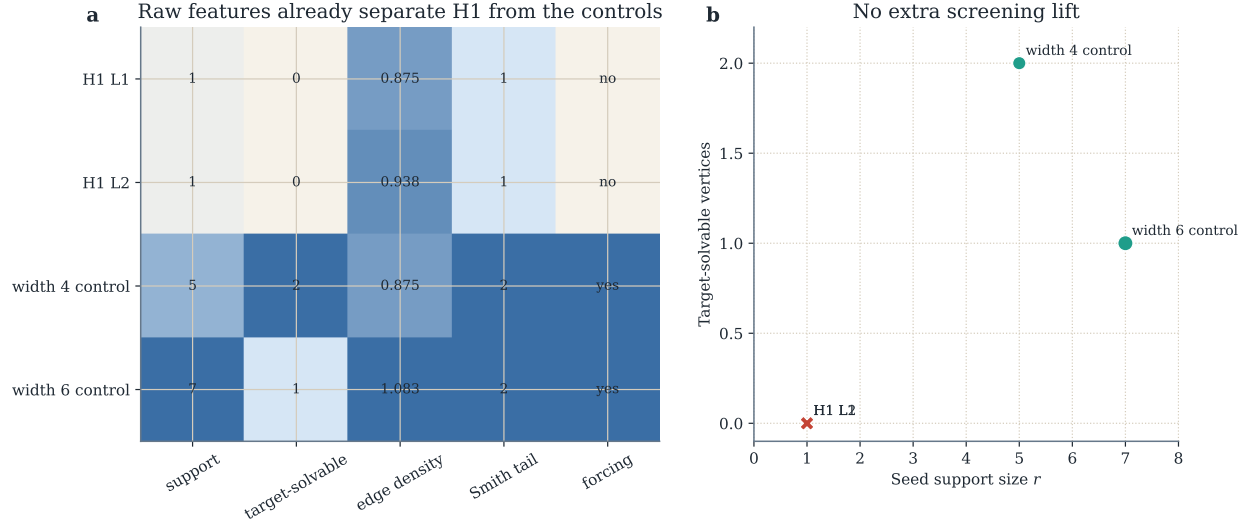


Figure 7: The saved H2 feature screen. **a**, normalized feature matrix on the four screened family IDs. **b**, support size and target-solvable count already separate the dead H1 rows from the surviving direct controls. The figure is useful precisely because of its narrow scope: it documents a local null result for the imported feature package, not a global no-go theorem.

8 Discussion

8.1 What the Repository Actually Established

At the current endpoint, the strongest scientifically safe identity of the project is the following.

- The exact verifier and compiler are understood for the low-height grid family actually used in the repo.
- The frozen H1 corridor pilot fails by a theorem-level one-seed obstruction.
- The passive-middle height-3 lift fails by a stronger row-quotient theorem.
- Every tested cellular-automaton-positive route beyond those theorems either fails before composition (H4, H5) or collapses to parity with matched direct search (H6, controller bank, metastability summaries).
- The best surviving exact certificates are direct low-height asymmetric width-4 rows at scores $13/7$ and $14/7$, still above the target 1.675 .

That is not the positive headline posed by the original user request. But it is a real contribution. The repo now contains exact obstructions, explicit direct certificates, matched ledger accounting, and a reproducible route audit that prevents dead cellular-automaton narratives from being reintroduced under softer language.

8.2 Limitations

The paper is still narrow in four important ways.

1. Every positive row lives in a low-height, low-width grid family with a frozen extractor. Nothing here implies a barrier for arbitrary constructible graphs.
2. The best saved exact score is $13/7 \approx 1.8571$, far above the target 1.675. The paper is a falsification study, not progress on the benchmark frontier itself.
3. The matched direct baselines in Phase 6 are much better than the Phase 4 corridor controls, but they are still local to the chosen families. There is no claim that $13/7$ is a natural or stable frontier outside those saved families.
4. The repo does not preserve hardware logs for the original search runs, so the paper intentionally makes no runtime-efficiency claims.

The saved Phase 4 ablations still matter here. They show that early corridor controls were boundary-sensitive and did not survive naive perturbations of R and T [Archivara Research Lab, 2026a]. That fragility motivated the later shell, defect-transport, and controller audits, but it does not by itself explain or refute the later $13/7$ direct rows.

8.3 Failure Modes and Open Questions

The route audits identify four concrete failure modes.

Seed bottlenecks. Theorem 13 shows that a unique seed direction with empty initial solved set is fatal. Future constructive routes must evade that obstruction without merely hiding the missing complexity inside the initial solved set.

Interface insufficiency. H4 demonstrates that a small frozen interface alphabet can fail before any scale-up comparison begins. This suggests that finite-state interface metaphors are much easier to propose than to justify in the exact arithmetic setting.

Quotient obstructions in higher geometry. Theorem 15 shows that moving to height 3 is not enough if the added row is passive. Any future higher-geometry route will have to escape the passive-middle template itself, not merely widen it.

Parity masquerading as mechanism. H6, controller enumeration, and metastability all exhibit the same danger: a local schedule or controller can look distinctive while still reproducing the best already-saved direct certificate. This is the precise sense in which the repo warns against mechanism inflation.

The natural open questions follow directly.

1. Is there any exact certificate family outside the passive-middle regime that genuinely exploits height > 2 without hiding the work in a larger extractor or a much larger X ?

2. Can direct low-height asymmetric search beat 13/7 while keeping the same level of hidden-complexity accounting used in the Phase 6 ledger?
3. What is the right exact benchmark protocol for broader constructible-graph grammars, where the convenience of the grid specialization disappears but the same exact verifier logic can still be preserved?
4. Can one formulate reusable obstruction theorems beyond one-seed and row-quotient arguments, rather than rediscovering them route by route?

8.4 Societal and Epistemic Implications

The direct societal impact of this work is low. Its broader value is methodological. Machine-assisted mathematical research is especially vulnerable to novelty inflation: a vivid metaphor, a finite-state simulator, or an imported vocabulary can look like progress even when the exact arithmetic object has not improved. The present paper is strongest precisely where it resists that inflation. It keeps theorem statements, exact computations, and local heuristic language on separate evidence boundaries. In this problem class, that discipline is not merely stylistic. It is part of the mathematical contribution.

9 Conclusion

The repository does not solve arithmetic Kakeya using cellular automata. What it does provide is sharper than an anecdotal failure. For the audited low-height grid families, the exact compiler and verifier are understood; the frozen H1 corridor pilot is ruled out by a one-seed theorem; the passive-middle height-3 lift is ruled out by a stronger row-quotient theorem; the H4, H5, and H6 route audits show no improvement over matched direct search; and the width-8 complexity ledger explains why the saved exploratory 29/14 row does not count as matched progress. The best surviving exact rows are direct low-height asymmetric certificates at scores 13/7 and 14/7, still above the target 1.675.

That is a negative result, but it is a reusable one. The paper leaves behind exact theorems, exact measurements, exact certificates, and a benchmark protocol that makes overclaiming much harder. For the present repository state, that is the honest scientific endpoint.

A Exact Certificates

This appendix reproduces the strongest saved positive certificates in the six-line verifier format.

A.1 asym_a Direct Certificate

```
13/7 m=10 r=3 n=8 t=1
[(0,0), (1,0), (0,1), (1,1), (2,-1)]
[2,4]
[{1:(0,1)}, {(1,1):(1,1), (1,2):(2,-1), (1,3):(1,0), (2,1):(1,1),
(2,2):(1,0), (2,3):(1,1)}]
```

```
[[1,4]]
[{{2,4}:(1,0)}, {{1,1}:(1,0)}, {{2,1}:(1,0)}]
```

A.2 asym_b Direct Certificate

```
13/7 m=10 r=3 n=8 t=1
[(0,0), (1,0), (0,1), (2,-1), (2,1)]
[2,4]
[{{1}:(0,1)}, {{(1,1):(2,-1)}, {(1,2):(2,1)}, {(1,3):(1,0)}, {(2,1):(2,-1)},
{(2,2):(1,0)}, {(2,3):(2,-1)}}]
[[1,4]]
[{{2,4}:(1,0)}, {{1,1}:(2,-1)}, {{1,1}:(1,0)}]
```

A.3 asym_c Direct Certificate

```
14/7 m=10 r=4 n=8 t=1
[(0,0), (1,0), (0,1), (1,2), (2,-1)]
[2,4]
[{{1}:(2,-1)}, {{(1,1):(1,0)}, {(1,2):(0,1)}, {(1,3):(1,0)}, {(2,1):(1,2)},
{(2,2):(1,0)}, {(2,3):(2,-1)}}]
[[1,1]]
[{{1,4}:(0,1)}, {{1,4}:(1,0)}, {{2,1}:(1,0)}, {{2,4}:(1,0)}]
```

A.4 Legacy Exploratory Width-8 Witness

```
29/14 m=20 r=9 n=16 t=2
[(0,0), (0,1), (1,0), (2,-1)]
[2,8]
[{{1}:(2,-1)}, {{(1,1):(1,0)}, {(1,2):(0,1)}, {(1,3):(1,0)}, {(1,4):(1,0)},
{(1,5):(1,0)}, {(1,6):(0,1)}, {(1,7):(1,0)}, {(2,1):(1,0)}, {(2,2):(2,-1)},
{(2,3):(2,-1)}, {(2,5):(1,0)}, {(2,6):(2,-1)}}]
[[1,1], [1,2]]
[{{1,6}:(0,1)}, {{2,1}:(0,1)}, {{1,5}:(0,1)}, {{2,5}:(0,1)},
{{1,8}:(0,1)}, {{2,3}:(0,1)}, {{2,4}:(0,1)}, {{2,7}:(0,1)}, {{2,8}:(0,1)}]
```

B Commands and Reproducibility

The saved artifacts record the exact commands used to produce the main route audits. We reproduce the central ones here.

```
python3 scripts/ca_kakeya_search.py h1-obstruction \
  --sum-value 1 --nonzero-count 4 \
  > results/phase4_h1_obstruction.json
```

```
python3 scripts/phase4_h2_screen.py \
```

```
--control results/phase4_width4_seed601.json \
--control results/phase4_width6_seed602.json \
--output results/phase4_h2_screen.json
```

```
python3 scripts/phase6_experiments.py typed-residue
python3 scripts/phase6_experiments.py complexity-frontier-aggregate
python3 scripts/phase6_experiments.py affine-shell
python3 scripts/phase6_experiments.py geometry-lift
python3 scripts/phase6_experiments.py defect-transport
python3 scripts/phase6_experiments.py boundary-controller
python3 scripts/phase6_experiments.py metastability

python3 scripts/generate_paper_figures.py
```

C Auxiliary Proof Detail

The completeness proof of Theorem 12 uses only one monotonicity fact, which we isolate here for convenience.

Lemma 17. *Let $S \subseteq S' \subseteq V$, and let $u \in V \setminus S'$. If u is solvable from S , then u is solvable from S' .*

Proof. If u is solvable from S , then there exists a relation whose support outside S is exactly $\{u\}$ and whose value at u is anti-diagonal. Enlarging the solved set from S to S' can only remove coordinates from the unsolved set, never add them. Therefore the same relation still has support outside S' equal to $\{u\}$ and remains a valid rule 2 witness for u . $\square$

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